Data validation and document retrieval

INSTRUCTOR ANSWERS - STAFF ONLY

Core questions 1-5

Use groups of five at suitable points in the session. Ask learners to explain their choice and one rejected option. Use the final five if helpful. Keep these group discussions separate from the existing individual knowledge-check scores.

1. Kestrel's contract already marks customer_id and customer_name as personal data. Why should it mark the history field as well?

Correct answer: D. Its notes are free text that can quote the customer, so they can contain any personal details.

History notes are free text, and some of them quote the customer, so a note can contain any personal detail that the customer gave. The contract therefore marks customer_id, customer_name, postcode_area and history as personal data. Whether a field is required is a separate rule: case_id is required but isn't personal data. The marking is also why a quality report that you share holds only each rejected record's identifier and reasons.

2. A colleague suggests that a model infer the case schema from the first record of each export, so that ingestion needs no written contract. Why does Kestrel keep a reviewed, versioned contract instead?

Correct answer: A. An inferred schema can change with the input and miss rules that one record doesn't show.

An inferred schema can vary with the input and omit requirements that the sample doesn't reveal, such as the UTC form of opened_at or a required summary. Kestrel's data owners review and version the contract, so every run applies a known set of rules and its decisions can be reproduced. A model may help draft the contract, so a ban on model involvement isn't the reason; the running job simply uses the approved version.

3. A document's header says Status: current, but its Last reviewed date is more than 365 days before the dataset's reference date. What does ingestion do with the document?

Correct answer: C. It flags the document as stale in the quality report, and it still loads it into curated/.

The freshness check flags documents older than 365 days, whatever their Status value, and it marks them for review rather than rejecting them. Stale and superseded documents still pass the schema and remain in curated/ with their metadata. Ingestion doesn't change the status either, because a step must never invent a value that the source didn't supply.

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4. A case record has a priority field that contract version 1.0 doesn't name, and its opened_at value is written day first. What does validate_case() return for the record?

Correct answer: B. It returns two reasons, because the model forbids extra fields and needs the exact UTC form.

CaseRecord sets extra="forbid", so a field that the contract doesn't name is an error rather than something to ignore. UtcInstant accepts only the exact ISO 8601 UTC form, so a date written day first fails instead of being read as something else. contract_problems() turns each validation error into one reason, so a record that breaks two rules gets two reasons, and the quality report lists both.

5. Halfway through a run, the team finds that the postcode_area rule rejects KT-10, a service area that Kestrel has just opened. What should happen?

Correct answer: B. The rule's owner publishes a new contract version, and ingestion runs again under that version.

A rule change is a decision for the field's owner, customer care, and it takes effect as a new contract version, so every run applies one known set of rules. Loosening the rule during the run would leave records accepted under version 1.0 that version 1.0 rejects, and nobody could tell which rules accepted them. Mapping or blanking the value would invent data that the source didn't supply.

Discussion: Ask the group to connect one of these ideas to today's Kestrel example. Use the Day 4 learner handbook to resolve uncertainty.

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Core questions 6-10

6. A developer proposes loading Kestrel's cases into the knowledge base as well, so that one store serves both kinds of question. What is the main risk?

Correct answer: D. A lookup would become a similarity search, which could return a case of a different customer.

A case is always read by its identifier, and DynamoDB returns the item whose partition key is case_id directly, with one GetItem call. In a knowledge base, the lookup becomes a similarity search, and the closest case can belong to a different customer, so similarity alone can't decide which case to return. The cases wouldn't skip the contract, because ingestion validates every record before it writes anything for a consumer to read.

7. The case export holds two records with the same case_id. Their summaries differ, and both pass every other rule. What does ingestion do?

Correct answer: C. It accepts the first record and rejects the second, because a person must decide which is right.

When a second record has the same case_id as an accepted one, ingestion compares the two. An identical copy counts as a duplicate and is written once, but a copy with different content is rejected, because choosing between two versions is a decision for a person. Ingestion has no rule that a later record is newer, and DynamoDB holds one item for each key.

8. A case arrives with no owner_team. Most Kestrel cases belong to customer care. What should ingestion do with the case?

Correct answer: B. Reject the case with its reason, because a guessed team could let the wrong staff read it.

The case lookup tool compares a case's owning team with the caller's team, so an invented owner_team decides who can read the case. The record goes to rejected/ with its reason, and the data owner supplies the correct value at the source. Recording the default in the quality report would make the guess visible, but it wouldn't make it correct, and a record that breaks the contract is never loaded.

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9. On a second run, put_item fails for one case with AccessDeniedException instead of ConditionalCheckFailedException. How does Kestrel's ingestion code treat the failure?

Correct answer: C. It raises the error, because only a failed condition counts as unchanged, not an access denial.

The code catches only ConditionalCheckFailedException, which DynamoDB returns when the stored checksum matches, and counts that refusal as unchanged. Every other error, such as an access denial, is raised, so that a real failure never looks like success. The rejected/ prefix is for records that broke the contract, and a permission problem says nothing about the record.

10. The instructor waits for an ingestion job with a limit of five seconds, and wait_for_ingestion() returns a record with the status TIMED_OUT. What does that record establish?

Correct answer: C. The job didn't finish within the chosen limit, and the job itself may still complete later.

TIMED_OUT means only that polling stopped at the deadline, and the record holds the job's identifier and the elapsed time. The limit ends the wait, not the AWS job, and in the demonstration a second wait on the same job returns COMPLETE. A job that fails is reported as failed at once, because the check raises an error when it sees the status FAILED.

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Core questions 11-15

11. An engineer traces a citation of kd_07. The manifest row's location matches the citation, but the SHA-256 checksum of the object there differs from the manifest. What can the engineer conclude?

Correct answer: A. The object isn't the version that ingestion accepted, so the citation doesn't establish the source.

The manifest records the SHA-256 checksum of each record as it was loaded, so a different checksum means that the bytes at that location aren't the version that ingestion accepted. If any step of the trace fails, the citation doesn't establish that the supplied source supports the answer. Overwriting the manifest's checksum would hide the difference instead of explaining it, and a sync reads objects without changing them.

12. A quality report shows 40 cases read, 36 accepted and 2 rejected, and it lists one identical duplicate. What should the engineer conclude?

Correct answer: D. One case is unaccounted for, so the ingestion code needs investigating before the results are used.

For cases, the number read should equal the accepted, rejected and identical duplicate records together, and here they make 39, not 40. A mismatch means that the report hasn't accounted for every input record, so the engineer investigates the code before using the results. A data quality score is a Glue Data Quality measure of rules passed, not a share of records, and the contract sets no tolerance for errors.

13. A Glue Data Quality evaluation of Kestrel's catalogued cases reports a data quality score of 80 per cent for a ruleset of five rules. What does the score mean?

Correct answer: A. Four of the five rules passed, which says nothing directly about how many cases are valid.

A data quality score is the percentage of the ruleset's rules that passed, so 80 per cent of five rules means that four passed. Kestrel's quality report counts records, which is a different measure. Identifying the records that failed, so that they can go to a separate location, is something the service does inside an ETL job, not in an evaluation of a catalogued table.

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14. A pipeline loads support tickets into a data lake every hour. It must keep invalid rows out of the lake and send them to a separate location. Where should the DQDL ruleset run?

Correct answer: B. Run it inside the Glue ETL job that loads the rows, which checks data in transit and can route failures.

Inside an ETL job, Glue Data Quality checks data in transit before the job loads it, and it can identify the records that failed, so that the job sends them elsewhere. A ruleset on a Data Catalog table measures data that is already stored, so it can't hold rows back before loading. CloudWatch and EventBridge receive the results of an evaluation; they don't evaluate rules.

15. Kestrel's refund and repair policies change several times a year, and staff must be able to check the source of every answer. Why does the course retrieve passages instead of fine-tuning a model on the documents?

Correct answer: A. Retrieval reads the current passages and their locations at question time, which fine-tuning lacks.

Retrieval supplies the current passages and their source locations when the question arrives, and after a document changes, ingestion and a sync update the searchable content. Fine-tuning alone gives neither a document-update process nor retrieved sources that a citation check can verify. A sync updates the index, not the model's weights, and fine-tuning adapts an existing model rather than training one from scratch.

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Core questions 16-20

16. A team wants retrieval to match small chunks but return the larger section around each match, so that the model gets more context. What does that require?

Correct answer: D. Use a customer-managed knowledge base with hierarchical chunking, set when its data source is created.

Hierarchical chunking keeps small child chunks inside larger parent chunks, and retrieval matches a child chunk and returns its parent. Only a customer-managed knowledge base offers it; a managed one offers Default, Fixed-size and No chunking, and its Default is fixed-size chunking of 300 tokens with 20 per cent overlap. The strategy is fixed when the data source is created, so an existing data source can't switch to it.

17. In the comparison, Retrieve returns five scored passages even for the five questions that no Kestrel document answers. What does this show about the scores?

Correct answer: D. Retrieve returns its top-ranked passages regardless, so the application decides what is relevant.

Retrieve returns the highest-ranked passages it finds, whether or not any of them answers the question, and a score only compares passages for one question. Deciding whether a passage is relevant enough is the application's job, and the course records these top scores so that Week 2 can choose a threshold. The measured scores overlap between the two groups of questions, so no single score marks a passage as an answer.

18. A developer calls RetrieveAndGenerate on Kestrel's knowledge base and receives "This operation is not supported for managed knowledge bases". Which approach does Kestrel use instead?

Correct answer: C. It calls Retrieve and writes the answer with its own Converse call, so its code checks each citation.

The API reference says that RetrieveAndGenerate can't be used with managed knowledge bases, which use AgenticRetrieveStream or Retrieve instead. Kestrel calls Retrieve and writes the answer with its own Converse call, so that its code can check every citation against the passages it supplied and the manifest. AgenticRetrieveStream generates a cited answer by default, but setting generateResponse to false turns the answer off.

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19. Week 2 will restrict a customer's retrieval to customer documents with a metadata filter on audience. Where should the value that the filter requires come from?

Correct answer: A. It should come from application code, which sets it from the caller's verified role on each call.

Access filters are built in application code from the caller's role and permissions. A filter that the model generates, or that relies on what the caller says about themselves, could include documents that the caller isn't authorised to read. The metadata file supplies each document's audience value for the filter to compare, but the value that the filter requires must come from the trusted caller context.

20. On a question that names the router's model, the keyword baseline ranks the supporting document first and the knowledge base ranks it third. On "my broadband keeps dropping", the reverse happens. What explains the difference?

Correct answer: B. Keyword ranking does well when the question shares the document's words, and hybrid search adds meaning.

BM25 scores each whole document by the words of the question that it contains, so it does well when the customer uses the document's own words, such as a model name. Hybrid search ranks by shared words and by closeness of meaning, so it can find the fault-reporting document for a question about a dropping connection. BM25 counts long documents for less, and the comparison runs both retrievers with no metadata filter, so that neither has an advantage.

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Optional extra questions 21-25

21. An engineer edits kd_15 in the source folder, but an hour later the assistant still cites the old wording. The data source has no sync schedule. What is the most likely cause?

Correct answer: A. The changed file hasn't been through ingestion and a sync, so the index holds the old chunks.

A knowledge base doesn't read a changed document until ingestion writes it to curated/ and its data source syncs, and without a schedule a sync runs only when someone starts one. A sync is incremental, but it processes every document that was added, changed or deleted, so a changed document is indexed again. The default schedule is on demand, not daily.

22. A client says that the assistant can trust whatever the knowledge base retrieves, because guardrails will be turned on for the model calls. Why is that reasoning unsafe?

Correct answer: B. Guardrails don't apply to passages retrieved from a knowledge base, which can still be wrong.

The Retrieve documentation states that guardrails apply to the input and the generated response, not to the passages retrieved from a knowledge base. A retrieved passage can therefore contain incorrect or out-of-date information even when guardrails are turned on. Kestrel validates the source data, records document status and checks each citation against the manifest and the supporting text.

23. An answer cites [kd_15] after each sentence, and kd_15 was among the passages supplied. One sentence states a figure that the passage doesn't contain. What does this case show?

Correct answer: B. The answer passes the citation check, which checks identifiers, so a review must catch the figure.

The citation check confirms that each cited identifier belongs to a supplied passage and a document in the manifest, and kd_15 meets both conditions. It doesn't compare a claim with the passage, so a person or an evaluation step must catch the unsupported sentence before a customer sees the answer. A citation tells you which passage the answer used; it doesn't prove that the passage supports the claim.

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24. A learner's AWS access through their employer will be ready at the start of Week 2. What can the learner do with Exercise 4 today?

Correct answer: D. Complete validate_case() and put_case_if_changed(), and run the check, which calls no AWS service.

The exercise's check tests the two functions without calling AWS, so a learner can complete and check them before they have an account. Only the cells that create the bucket and the table and ingest the data need the learner's own access, which learners arrange, through their employer or a personal account, by the start of Week 2. The plan makes no AWS call, but it doesn't simulate the writes, and the course account is used for the instructor's demonstration.

25. A colleague creates a knowledge base whose role can read curated/documents/ only. Its first sync returns COMPLETE, but its statistics show no documents indexed. What should the engineer conclude?

Correct answer: C. The role must also read curated/metadata/, and COMPLETE doesn't mean that anything was indexed.

The knowledge base's role must list the bucket and read the objects under curated/, which holds both the documents and their metadata files. With access to curated/documents/ alone, a sync can finish as COMPLETE with a warning about permission to list objects and index nothing. sync() returns the job's failure reasons as warnings, so the engineer reads the warnings and the indexing counts as well as the status.

Discussion: Ask the group to connect one of these ideas to today's Kestrel example. Use the Day 4 learner handbook to resolve uncertainty.